Model 型号	LF100LA	Specification No. 规格书编号	PBRI-LF100L-V2.1-D06-01	Version 版本	C

Product Specification
产品规格书

Prismatic LFP Cell
方形磷酸铁锂电池

Model: LF100LA
型号: LF100LA

Drafted by 编制	Product Design Checked by 产品设计审核	Quality Checked by 品质审核	Sales Checked by 销售审核	Approved by 批准
宋家新	丁静	张利霞	刘小亚	丁丁

Customer Recipient 客户接收栏
Company Name 公司名称:
Approved by 批 准:
Date 日 期:

Oct. 2024
EVE Power Co., Ltd.
湖北亿纬动力有限公司

Model 型号	LF100LA	Specification No. 规格书编号	PBRI-LF100L-V2.1-D06-01	Version 版本	C

Customer Requirements 客户要求

Customers shall communicate and reach an agreement with EVE Power Co., Ltd. regarding special requirements beyond the product specifications. If certain applications and operation conditions are out of the description of this specification, EVE may design and manufacture products according to customer's inputs.

客户输出产品规格书以外的特殊需求信息需与亿纬动力沟通并达成一致。例如客户有一些特别的应用或者操作条件不同于此文件中所描述的，亿纬动力根据客户的特别要求进行产品的设计和生产。

No. 序号	Special Requirements 特殊要求	Standards 标准
1		
2		
3		
4		
5		

Customer Code 客户代码： _____ **Signature 签字：** _____ **Date 日期：** _____

Model 型号	LF100LA	Specification No. 规格书编号	PBRI-LF100L-V2.1-D06-01	Version 版本	C

Revision History 变更履历

Version 版本	Date 日期	Contents 更改内容	Checked By 确认人
A	2022.05.20	First issue 新版发行	Peng Chen 陈鹏
B	2022.09.20	1. Capacity calibration, the current is adjusted from 0.5C to 0.2C; 2. The calibration capacity is adjusted from the minimum 100Ah to 102Ah. 1、容量标定，电流由 0.5C 调整为 0.2C； 2、标定容量由最低 100Ah 调整为 102Ah。	Ming Chen 陈明
C	2024.10.08	1. Adjust chapter content according to the latest EVE specification template. 1、按照 EVE 最新规格书模板，调整章节内容。	Jiaxin Zhu 朱家新

Model 型号	LF100LA	Specification No. 规格书编号	PBRI-LF100L-V2.1-D06-01	Version 版本	C

Contents

目录

CUSTOMER REQUIREMENTS 客户要求 ..... **I**

REVISION HISTORY 变更履历 ..... **II**

TERM DEFINITION 术语定义 ..... **IV**

1 SCOPE OF APPLICATION 适用范围 ..... **- 1 -**

2 CELL SPECIFICATIONS 电池规格 ..... **- 1 -**

2.1 Product Specifications 产品规格 ..... - 1 -

2.2 Electrical Performance 电性能 ..... - 2 -

2.3 Charging Parameters 充电参数 ..... - 4 -

2.4 Discharging Parameters 放电参数 ..... - 5 -

2.5 Safety Performance 安全性能 ..... - 6 -

3 PRODUCT END-LIFE MANAGEMENT 产品寿命终止管理 ..... **- 7 -**

4 APPLICATION CONDITIONS 应用条件 ..... **- 7 -**

5 SAFETY PRECAUTIONS 安全防范 ..... **- 13 -**

6 DISCLAIMER 免责声明 ..... **- 15 -**

7 RISK WARNING 风险警告 ..... **- 16 -**

8 OTHERS 其他 ..... **- 16 -**

9 CONTACT INFORMATION 联系方式 ..... **- 18 -**

APPENDIX 1: CELL DRAWING OF LF100LA 附录 1：LF100LA 电池图纸 ..... **- 19 -**

APPENDIX 2: TEST CONDITIONS AND METHODS 附录 2：试验条件及方法 ..... **- 20 -**

Model 型号	LF100LA	Specification No. 规格书编号	PBRI-LF100L-V2.1-D06-01	Version 版本	C

Term Definition 术语定义

Product 产品:

Refers to the LF100LA rechargeable lithium-ion cell with prismatic aluminum shell manufactured by EVE Power Co., Ltd. (hereinafter referred to as EVE) in this specification.

本规格书中的产品是指湖北亿纬动力有限公司（以下简称 EVE）生产的 LF100LA 方形铝壳锂离子电池。

Customer 客户:

Refers to the buyer in the product sales contract signed with EVE.

指与 EVE 签署产品销售合同中的买方。

Environment temperature 环境温度:

The ambient temperature where the cell is located.

电池所处的周围环境温度。

Cell temperature 电池温度:

The temperature measured by temperature sensor installed at the center of cell surface. The selection of temperature sensor and measuring line shall be jointly agreed by EVE and the customer.

由接入电池表面中心的温度传感器测量的电池表面的温度，温度传感器和测量线路的选择由 EVE 和客户共同商定。

Fresh cell 新鲜电池:

Refers to the cell within 7 days from the customer's receipt date (for domestic transportation only)

指客户收货日期算起 7 天以内的电池（仅限国内运输）。

Rate 倍率:

The ratio of the charging or discharging current to the capacity which measured by the battery management system. For example, if the cell capacity is 100 Ah and the charging or discharging current is 50 A, the charging or discharging rate is 0.5C. If the cell capacity drops to 80 Ah and the charging or discharging current is 40 A, the charging or discharging rate is 0.5C.

充/放电电流与电池管理系统监测的电池的容量值的比率。例如：电池容量为 100 Ah，充电或放电电流为 50A 时，则充电或放电倍率为 0.5C；当电池容量跌落为 80 Ah，充电或放电倍率为 40 A 时，则充电或放电倍率为 0.5C。

State of charge 荷电状态:

Under unloaded conditions, the ratio of the cell capacity state to the nominal capacity measured in ampere-hour or watt- hour. The abbreviation is expressed by SOC. For example, if the capacity at 100 Ah is 100% SOC, the capacity at 0 Ah is 0% SOC.

在无负载的情况下，以安培小时或者以瓦特小时为单位计量的电池容量状态与标称容量的比值，缩写用 SOC

表示。例如：若将容量为 100 Ah 的状态视为 100% SOC，若容量为 0 Ah 时，SOC 为 0%。

State of health 健康状态:

The ratio of actual cell capacity to nominal capacity, abbreviated as SOH. For example, if the cell capacity of 100 Ah is 100% SOH, when the cell capacity decays to 80 Ah, it's at 80% SOH.

电池实际容量与标称容量的比值，缩写用 SOH 表示。例如：电池容量 100 Ah 为 100% SOH，容量衰减为 80 Ah 时，SOH 为 80%。

Cycle 循环:

A cycle means the cell being charged and discharged once according to the charging and discharging standards. The cell shall be charged and discharged once according to the specified charging and discharging standards as a cycle. The cycle includes short-term normal charging or a combination of regenerative charging and discharging processes. In the charging process, sometimes there is only normal charging and no regenerative charging. The discharge can be formed by combining some partial discharges.

电池按规定的充放标准充放一次为一个循环。循环包括短时的正常充电或者再生充电和放电过程的组合，在充电过程中有时只有正常充电而无再生充电的情况。放电可以由一些部分放电组合在一起形成。

Open circuit voltage 开路电压:

The voltage of the cell measured when unloaded or circuit is disconnected. The abbreviation is expressed by OCV.

没有接入任何负载和电路时测得的电池的电压，缩写用 OCV 表示。

AC resistance 交流内阻:

Apply 1kHz sine wave current between the positive and negative poles of the cell, and the internal resistance obtained is AC resistance, which is abbreviated as ACR. The test method is described in appendix 2:1.6 of this specification.

给电池正负极注入 1kHz 的正弦波电流，测试所得到的内阻值，缩写用 ACR 表示，测试方法如本规格书附录 2 第 1.6 条所述。

DC resistance 直流内阻:

The ratio of the voltage changes to the corresponding current change under working conditions, and the abbreviation is DCR. The test method is as described in appendix 2:1.6 of this specification.

工作条件下电池的电压变化与相应的电流变化之比，缩写用 DCR 表示，测试方法如本规格书附录 2 第 1.6 条所述。

Module 模组:

The intermediate product between single cell and pack, which is formed by lithium-ion cells in series and parallel after installing cell monitors and management devices.

锂离子电池经串并联方式组合，加装单体电池监控与管理装置后形成的电池与 pack 的中间产品。

Pulse current 脉冲电流:

Model 型号	LF100LA	Specification No. 规格书编号	PBRI-LF100L-V2.1-D06-01	Version 版本	C

The currents that appear periodically are called pulse currents. The pulse currents appear either in the same direction or in alternating positive and negative directions.

以周期重复出现的电流为脉冲电流，脉冲电流或是以同一方向出现，或是以正、负交替变换方向出现。

Compression force 压缩力:

When the module is assembled, the cell can withstand the force perpendicular to the cell stacking direction.

模组组装时，电池可承受压缩力的安全边界。

Swelling force 膨胀力:

The force on the clamp due to cell expansion during use, which may caused by inherent characteristic changes, such as the rebound of electrode thickness.

在使用过程中，因极片厚度反弹等固有特性导致电池膨胀，而对约束夹具产生的作用力。

Units of measurement： Refer to following table

测量单位： 见下表

Table 1 Units of Measurement

表 1 测量单位

No. 序号	Units 单位	Abbreviation 简写	Type of units 单位类型
1	Volt 伏特	V	Voltage 电压单位
2	Ampere 安培	A	Current 电流单位
3	Ampere-Hour 安培-小时	Ah	Capacity 容量单位
4	Watt-Hour 瓦特-小时	Wh	Energy 能量单位
5	Ohm 欧姆	Ω	Resistance 电阻单位
6	Milliohm 毫欧姆	mΩ	Resistance 电阻单位
7	Degree Celsius 摄氏度	℃	Temperature 温度单位
8	Millimeter 毫米	mm	Length 长度单位
9	Second 秒	s	Time 时间单位
10	Hertz 赫兹	Hz	Frequency 频率单位
11	Newton 牛顿	N	Force 力单位
12	Kilogram-Force 千克力	kgf	Force 力单位

Model 型号	LF100LA	Specification No. 规格书编号	PBRI-LF100L-V2.1-D06-01	Version 版本	C

1 Scope of Application 适用范围

This document describes the specification of the LF100LA lithium-ion cell manufactured by EVE Power Co., Ltd.

本产品规格书适用于 EVE 生产的 LF100LA 型号锂离子电池。

2 Cell Specifications 电池规格

2.1 Product Specifications 产品规格

Table 2 Product Specifications

表 2 产品规格

Items 项目		Specifications 规格	Remarks 备注
Nominal Capacity 标称容量		102 Ah	0.2C / 0.2C, 25℃ ± 2℃, 2.5 V ~ 3.65 V Fresh cell 新鲜电池
Nominal Energy 标称能量		326.4 Wh	
Nominal Voltage 标称电压		3.2 V	/
End-of-charge Voltage (U_{max}) 充电限制电压 (U_{max})		3.65 V	/
End-of-discharge Voltage (U_{min}) 放电截止电压 (U_{min})		2.5 V（T >0℃） 2.0 V（T ≤0℃）	/
Standard Charging Current 标准充电电流		0.2C	25℃ ± 2℃
Max. Continuous Charging Current 最大持续充电电流		1.0C	25℃ ± 2℃
Standard Discharging Current 标准放电电流		0.2C	25℃ ± 2℃
Max. Continuous Discharging Current 最大持续放电电流		1.0C	25℃ ± 2℃
Initial Internal Resistance 初始内阻		0.26±0.10 mΩ	AC, 1 kHz, Delivery SOC, Fresh cell 新鲜电池，出货 SOC
Weight 重量		1985 g ±100 g	/
Dimensions (With	Height1 高度 1（H1）	118.5 mm ±0.5 mm	With Terminal 含极柱

Model 型号	LF100LA	Specification No. 规格书编号	PBRI-LF100L-V2.1-D06-01	Version 版本	C

insulation film) 尺寸 （含蓝膜）	Height2 高度 2（H2）	115.7 mm ±0.5 mm	Without Terminal 不包含极柱
	Length 宽度（L）	160.0 mm±0.8 mm	/
	Thickness 厚度（T）	50.1 mm±0.5mm	(200 kgf ±20 kgf compression force, Delivery SOC) （200 kgf ±20 kgf压缩力，出货 SOC）
	Center distance between the poles 极柱中心距（D）	97.0 mm ±0.3 mm	/
Operation Temperature 工作温度	Charging Temperature 充电温度	0℃ ~ 65℃	/
	Discharging Temperature 放电温度	-20℃ ~ 65℃	/
Storage Temperature 存储温度	1 year 1 年	0℃ ~ 35℃	Delivery SOC status 出货 SOC 状态
	1 month 1 个月	-20℃ ~ 45℃	
Self-discharge 自放电	First Month 首月	≤3.0%	Delivery SOC status, 25℃ ±2℃ storage 出货 SOC 状态，25℃±2℃ 储存
	After First Month 首月后	≤1.5% / M	

2.2 Electrical Performance 电性能

Table 3 Electrical Performance Parameters

表 3 电性能参数

Items 项目	Specifications 规格			Testing Methods 测试方法
Capacity 容量 （25℃）	Items / Rate	Discharging capacity 放电容量		Appendix 2: 1.9 附录 2: 1.9
	0.2C	C_0≥102Ah		
Rate Discharge Performance 倍率放电性能 （25℃）	Items / Rate	Discharging capacity 放电容量	Capacity Retention 容量保持率	Appendix 2: 1.11 附录 2: 1.11
	0.5C	C_1	C_1/ C_2 ≥100%	
	1.0C	C_2	C_2/ C_2 =100%	
	2.0C	C_3	C_3/ C_2 ≥98%	
	2.5C	C_4	C_4/ C_2 ≥95%	

Model 型号	LF100LA	Specification No. 规格书编号	PBRI-LF100L-V2.1-D06-01	Version 版本	C

High/Low Temperature Discharge Performance 高/低温放电性能	Items / Temp.	Discharging capacity 放电容量	Capacity Retention 容量保持率	Appendix 2: 1.10 附录 2: 1.10
	-20℃	C_5	$C_5/C_1 \geq 65\%$	
	0℃	C_6	$C_6/C_1 \geq 80\%$	
	10℃	C_7	$C_7/C_1 \geq 85\%$	
	55℃	C_8	$C_8/C_1 \geq 100\%$	
DCR 直流内阻	25℃_0.2C_30s & 1C_5s @50%SOC	0.8±0.3mΩ		Appendix 2: 1.6 附录 2: 1.6
Energy Efficiency 能量效率	25℃, 0.5C Energy Efficiency 25℃，0.5C 能量效率	η≥94%		Appendix 2: 1.12 附录 2: 1.12
Cell temperature rise without wind 电芯无风温升	35℃, 1C/1C, discharge 35℃，1C/1C 放电	≤15℃		Temperature rise refers to the difference of the cell surface temperature before and after discharging 温升是指放电前后的电池表面温度差值 Appendix 2: 1.18 附录 2: 1.18
Floating Charge 浮充性能	25℃ 15 Years 25℃ 浮充 15 年	≥70%SOH		Appendix 2: 1.13 附录 2: 1.13
Capacity Retention and Recovery 荷电保持与恢复 （100% SOC）	Items / Temp.	Capacity recovery 容量恢复率		Appendix 2: 1.15 附录 2: 1.15
	25℃ & 28 days	≥ 97%*C_0		
	45℃ & 28 days	≥ 96%*C_0		
Cycle Life 循环寿命	25℃ 0.5C Cycle 25℃ 0.5C 循环	0.5C/0.5C 5000 cycles, 80% SOH		Appendix 2: 1.14 附录 2: 1.14
	25℃ 1C Cycle 25℃ 1C 循环	1C/1C 4000 cycles, 80% SOH		
	45℃ 0.5C Cycle 45℃ 0.5C 循环	0.5C/0.5C 2000 cycles, 80% SOH		
	45℃ Cycle 45℃ 1C 循环	1C/1C 1800 cycles, 80% SOH		
Storage 存储性能 （50% SOC）	Items / Temp.	Discharging capacity recovery 放电容量恢复率		Appendix 2: 1.16 附录 2: 1.16
	25℃ & 28days	≥ 98%*C_0		

	45℃ & 28days	≥ 97%*C_0	
Swelling Force 膨胀力	70% SOH	≤ 25000 N	Appendix 2: 1.17 附录 2: 1.17

2.3 Charging Parameters 充电参数

2.3.1 Charging Mode 充电模式

Table 4 Charging Mode Parameters

表 4 充电模式参数

Parameters 参数	Specifications 规格	Conditions 条件
Standard charging current 标准充电电流	0.2C	25℃ ±2℃
Maximum continuous charging current 最大充电可持续电流	1.0C	25℃ ±2℃
Standard charging voltage 标准充电电压	Single cell ≤ 3.65 V 单体电池 ≤ 3.65 V	
Standard charging mode 标准充电模式	Charge to 3.65 V with a constant current of 0.2C, and then transfers to constant voltage charging at 3.65V until the charging current is less than or equal to 0.05C 以 0.2C 恒流恒压充电至 3.65 V，截止电流 0.05C	
Standard charging temperature 标准充电温度	25℃ ±2℃	
Absolute charging temperature (cell temperature) 绝对充电温度（电池温度）	0℃ ~ 65℃	No matter what charging mode the cell is in, once the cell temperature exceeds the absolute charging temperature range, stop charging. 无论电池处于何种充电模式，电池温度一旦超过绝对充电温度范围，即停止充电。
Absolute charging voltage 绝对充电电压	Max 3.65 V 最大 3.65V	No matter what charging mode the cell is in, once the cell voltage exceeds the absolute charging voltage, stop charging. 无论电池处于何种充电模式，电池电压一旦超过绝对充电电压，即停止充电。

2.3.2 Other Charging Modes 其他充电模式

Table 5 Continuous Charging Modes (unit: C-Rate)

表 5 持续充电模式（单位：C-Rate）

Cell temperature /℃ **电池温度**/℃		(-10~0]	(0~5]	(5~10]	(10~15]	(15~20]	(20~25]	(25~60]	65
SOC	0%~80%	0.05	0.1	0.2	0.4	0.5	0.6	1.0	0
SOC	80%-90%	0	0.06	0.2	0.2	0.3	0.6	1.0	0
SOC	90%-100%	0	0.06	0.1	0.1	0.1	0.6	1.0	0

2.4 Discharging Parameters 放电参数

2.4.1 Discharging Mode 放电模式

Table 6 Discharging Mode Parameter

表 6 放电模式参数

Parameters 参数	Product specifications 产品规格	Conditions 条件
Standard discharging current 标准放电电流	0.2C	25℃ ±2℃
Maximum continuous discharging current 最大放电可持续电流	1.0C	25℃ ±2℃
Standard discharge mode 标准放电模式	Discharge to 2.5 V with a constant current of 0.2C 以 0.2C 恒流放电至 2.5 V	
Discharge cut-off voltage 放电截止电压	2.5 V	Temperature T > 0℃ 温度 T > 0℃
	2.0 V	Temperature T ≤ 0℃ 温度 T ≤ 0℃
Standard discharging temperature 标准放电温度	25℃ ±2℃	
Absolute discharging temperature (cell temperature) 绝对放电温度（电池温度）	-20℃ ~ 65℃	No matter what discharging mode the cell is on, once the cell temperature exceeds the absolute discharge temperature range, stop discharging. 无论电池处于何种放电模式，电池温度一旦超出绝对放电温度范

Model 型号	LF100LA	Specification No. 规格书编号	PBRI-LF100L-V2.1-D06-01	Version 版本	C

		围，即停止放电。
Absolute discharging voltage 绝对放电电压	Min 2.0 V 最小 2.0 V	No matter what kind of discharging mode the cell is on, once the cell voltage is less than the absolute discharge voltage, stop discharging. 无论电池处于何种放电模式，电池电压一旦小于绝对放电电压，即停止放电。

2.4.2 Other Discharging Modes 其他放电模式

Table 7 Continuous Discharging Rate (unit: C-Rate)

表 7 持续放电倍率（单位：C-Rate）

Cell temperature /℃ **电池温度**/℃		-20	-10	0	10	20	25	30	35	40	45	50	55	60	65
SOC	40% ~ 100%	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.5	0.5	0.5	0
SOC	40%-30%	0.5	0.5	0.5	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.5	0.5	0.5	0
SOC	20%-30%	0.2	0.3	0.3	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.5	0.5	0.5	0
SOC	10%-20%	0.1	0.2	0.3	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.5	0.5	0.5	0
SOC	0%-10%	0.1	0.1	0.1	0.5	1.0	1.0	1.0	0.5	0.5	0.5	0.5	0.5	0.5	0

2.5 Safety Performance 安全性能

Table 8 Safety Performance Parameters

表 8 安全性能参数

Items 项目	Specifications 规格	Test Methods 测试方法
Over-charge 过充电	No fire, No explosion 不起火、不爆炸	Appendix 2: 1.19.1 附录2: 1.19.1
Over-discharge 过放电	No fire, No explosion 不起火、不爆炸	Appendix 2: 1.19.2 附录 2: 1.19.2
External Short-circuit 外部短路	No fire, No explosion 不起火、不爆炸	Appendix 2: 1.19.3 附录 2: 1.19.3
Crush Test 挤压	No fire, No explosion 不起火、不爆炸	Appendix 2: 1.19.4 附录 2: 1.19.4
Drop Test 跌落	No fire, No explosion 不起火、不爆炸	Appendix 2: 1.19.5 附录 2: 1.19.5

Model 型号	LF100LA	Specification No. 规格书编号	PBRI-LF100L-V2.1-D06-01	Version 版本	C

Low Pressure 低气压	No fire, No explosion, No leakage, No venting, No seepage, and the change in OCV is less than 10% of the pre-test OCV. 不起火、不爆炸、不漏液、不排气、不泄漏、OCV 变化<10%测试前 OCV	Appendix 2: 1.19.6 附录 2: 1.19.6
Thermal Runaway 热失控	No fire, No explosion 不起火、不爆炸	Appendix 2: 1.19.7 附录 2: 1.19.7
Nail penetration 针刺	No fire, No explosion 不起火、不爆炸	Appendix 2: 1.19.8 附录 2: 1.19.8

Notes 备注:

The descriptions above only represent compliance at the cell level. The safety compliance at the module and the system level requires the customer to perform design verification.

以上仅代表电芯层级的符合性。模组或系统层级的安全符合性，需要客户设计验证。

3 Product End-life Management 产品寿命终止管理

The cell life is limited. Customers should establish an effective tracing system to monitor and record the internal resistance and capacity of each cell during its life. The measurement method and calculation method of internal resistance and capacity need to be discussed and agreed between the customer and EVE. When the internal resistance of the cell in use exceeds 150% of the initial internal resistance, or the capacity is less than 70% of nominal capacity, the cell should not to be used any more. Violation of this requirement will exempt EVE from its responsibility for product quality assurance in accordance with the product sales agreement and this specification, as well as all related liabilities such as loss compensation caused thereby.

电池使用期限是有限的，客户应建立有效的跟踪系统监测并记录每个使用期限内电池的内阻和容量。内阻及容量的测量方法和计算方法需要客户和 EVE 共同讨论和双方同意。当使用中电池的内阻超过这个电池最初内阻的150%或容量小于标称容量的 70%，应停止使用电池。违反该项要求，将免除 EVE 依据产品销售协议以及本规格书所应承担的产品质量保证责任及由此引起的损失赔偿等一切相关责任。

4 Application Conditions 应用条件

Customer shall ensure strict compliance with the following cell application conditions:

客户应当确保严格遵守以下与电池相关的应用条件

4.1 Customer shall configure a battery management and monitoring system to strictly monitor, manage and protect each cell, and provide detailed information of the BMS, including but not limited to its design, features, setting, and data file format to EVE for design review and record keeping. And a battery management archive shall be established to keep all monitoring data of the cells, so as to be a reference for problems tracing and product quality responsibility division. EVE is not responsible for product quality assurance if no complete monitoring data of the battery system during its service life is provided.

客户应配置电池管理和监控系统，严密监控、管理与保护每个电池，并向 EVE 提供电池管理系统详细的设计方案、系统特点、框架、系统数据等相关信息，以供 EVE 对该系统进行设计评估，并建立电池管理档案，保存完整的电池运转的监测数据，用做问题追溯及产品质量责任划分的参考。不具备完整的电池系统使用期限内的监测数据的，EVE 不承担产品质量保证责任。

4.2 Waterproof and dustproof problems shall be fully considered in the pack design, and the pack must meet the waterproof and dustproof grade stipulated by relevant national standards. EVE is not responsible for the damage (such as corrosion, rust, etc.) of the cell caused by waterproof and dustproof problems.

电池包设计中应充分考虑电池的防水、防尘问题，电池包须满足国家有关标准规定的防水、防尘等级。由于防水、防尘问题而导致的电池的损坏（如腐蚀、生锈、等），EVE 不承担质量保证责任。

4.3 It is forbidden to mix different types of cells in the same battery system, otherwise, EVE will not be responsible for the quality assurance.

禁止不同型号电池在同一电池系统中混用，否则，EVE 不承担质量保证责任。

4.4 The design of the BMS shall meet the safety voltage and operating temperature limitations in Table 9 and Table 10.

电池管理系统设计需满足表 9 和表 10 的安全电压和操作温度限制条件

Table 9 Safety Limit Voltage Parameters

表 9 安全限制电压参数

Items 项目	**Categories** 类别	**Parameters** 参数	**Protective Actions** 保护动作
Charging Voltage 充电电压	Charging Ends 充电终止	3.65 V	When the cell voltage reaches 3.65 V, stop charging. 当电池电压达到 3.65 V 时终止充电。
	First Over-Charging Protection 第一级过充保护	3.70 V	Decrease current or power 降流或降功率
	Second Over-Charging Protection	3.80 V	Stop charging 停止充电

	第二级过充保护		
Discharging Voltage 放电电压	Discharging Ends 放电终止	2.5 V (> 0℃) 2.0 V (≤ 0℃)	When the cell voltage reaches 2.5 V (0℃) or 2.0 V (≤0℃), stop charging. 当电池电压达到 2.5 V (> 0℃) 或者 2.0 V (≤ 0℃) 时终止充电。
	First Over-Discharging Protection 第一级过放保护	1.90V	Decrease current or power 降流或降功率
	Second Over-Discharging Protection 第二级过放保护	1.80V	Stop discharging 停止放电
BMS protection BMS 保护	Short circuit protection 短路保护	Short circuit is not allowed 不允许短路	When a short circuit occurs, the overcurrent protection device will disconnect the cell. 发生短路时，由过流器断开电池
Over current protection 过流保护	Reference 2.3 & 2.4 参考第 2.3 和 2.4 条		BMS controls the charging/discharging current within specifications. 电池管理系统控制充放电电流符合规格

Table 10 Safety Limit Temperature Parameters

表 10 安全限制温度参数

Items 项目	Parameters 参数	Remarks 备注
Recommended operating temperature range 推荐操作温度范围	10℃ ~ 45℃	Recommended cell using temperature range 推荐使用电池的温度范围
Maximum operating temperature 最高操作温度	65℃	If the cell temperature exceeds the maximum operating temperature, the cell current needs to be reduced to 0. 如果电池使用温度超过最高操作温度，电流需要降为 0。

Model 型号	LF100LA	Specification No. 规格书编号	PBRI-LF100L-V2.1-D06-01	Version 版本	C

Minimum operating temperature 最低操作温度	-30ºC	If the cell temperature exceeds the minimum operating temperature, the cell current needs to be reduced to 0. 如果电池使用温度低于最低操作温度，电流需要降为 0。
Maximum safe temperature 最高安全温度	65℃	If the battery temperature exceeds the maximum safe temperature, it will cause irreversible and permanent damage to the battery, and the user should not use it higher than the maximum safe temperature. 如果电池使用温度超过最高安全温度，将会造成电池不可逆的永久性损坏，用户使用时不得高于最高安全温度。
Minimum safe temperature 最低安全温度	-30℃	If the battery temperature exceeds the minimum safe temperature, it will cause irreversible and permanent damage to the battery, and the user should not lower the minimum safe temperature when using it. 如果电池使用温度超过最低安全温度，将会造成电池不可逆的永久性损坏，用户使用时不得低于最低安全温度。
Over heat protection 过热保护	Reference 2.3 & 2.4 参考第 2.3 和 2.4 条	Stop charging/discharging when the temperature exceeds the limitation in this specification 当温度超过本规格书规定时，终止充电/放电

Notes 备注：

a) If the cell charging voltage exceeds the cut-off voltage, corresponding protective actions need to be taken. If the cell discharging voltage reaches the cut-off voltage, it is necessary to charge as soon as possible to prevent it from being over-discharged. EVE shall not be responsible for any cell quality issues caused by over-discharge and exceeding the protection voltage.

电池充电超过终止电压时，需依据相应的保护动作采取措施。当电池放电电压到达终止电压时，需尽快充电，防止电池进入过放状态。对于超出保护电压和电池过放导致的电池质量问题，EVE 不承担任何保证责任。

b) It's prohibited to charge the cell at low temperatures (lower than 0℃) and the minimum safety temperature of this specification, otherwise EVE will not be responsible for any quality assurance liability. The heat dissipation design of pack may affect cell electrical performance, EVE will not be responsible for any liability regarding cell quality issues caused by the pack heat dissipation design.

电池禁止在低温下（0℃以下）充电及在规定的最低安全温度下放电，否则 EVE 不承担任何电池质量保证责任。电池包的散热设计会影响电池性能，因电池包散热设计问题导致的电池质量问题，EVE 不承担任何责任。

4.5 Recommendations for Module Welding Parameters 模组焊接参数建议

Table11 Welding Parameters

表 11 焊接参数表

Items 项目		Specifications 规格	Remarks 备注
Welding Parameter of Al Busbar 铝巴焊接参数	Pole material 极柱材质	Al	/
	Laser Welding Depth 激光焊接熔深	≤ 2.0 mm	/
	Max Pressure on Poles 极柱承受最大压力	500 N	The maximum force on the poles in vertical direction with no deformation. 极柱承受最大垂直力，不发生变形
	Max Torque on Poles 极柱承受最大扭矩	4 N·m	The maximum torsion on the poles with no loosen. 极柱承受最大扭曲，不松动
	Max Temperature of Poles 极柱承受最大温度	130℃	The maximum temperature that the poles bear before the plastic pad deforms. 极柱承受最大温度，塑胶垫不发生变形
	Pole center distance 极柱中心距	97mm	/

4.6 Cell Compression Force 电池压缩力

Table 12 Cell Compression Force Limit Parameters

表 12 电池压缩力限制参数

Items 项目	Compression Force 压缩力
Recommended compression force 推荐压缩力	1000 N ~ 4000 N
Instantaneous maximum compression force 瞬时最大压缩力	≤ 8000 N

4.7 Cell Swelling Force 电池膨胀力

The inherent characteristics such as the rebound of the electrode thickness may lead to cell expansion during use which

generates force on the clamp, and the force increases with the attenuation of the cell capacity. The cell swelling force at BOL and EOL (70% SOH) refer to Table 3.

在使用过程中，因极片厚度反弹等固有特性导致电池膨胀，而对约束夹具产生的作用力；电池的膨胀力随着电池容量的衰减而增加，电池 BOL 和 EOL（70% SOH）膨胀力参考表 3。

4.8 Recommended Temperature Control 温度管理建议

The recommended temperature collection points are the poles and the code when collecting temperature of the cell surface. The cell thermodynamic parameters needed in the thermal management system are shown in the following table.

对电池表面进行温度采集时，建议温度采集点布置在极柱或刻码。对热管理系统中所需的电芯热力学参数见下表：

Table 13 Cell Thermal Conductivity Parameters

表 13 电池导热系数参数

Mean thermal conductivity 导热系数均值	Thermal Conductivity W/(m K) 导热系数 W/(m K)		
	X direction 方向	Y direction 方向	Z direction 方向
	12.2~ 14.2 W/(m K)	3.2 ~ 4.2 W/(m K)	8.6 ~ 10.6 W/(m K)
Mean heat capacity 热容均值	Heat Capacity 热容 (kJ/(kg K))		
	0.9 ~ 1.1 kJ/(kg K)		

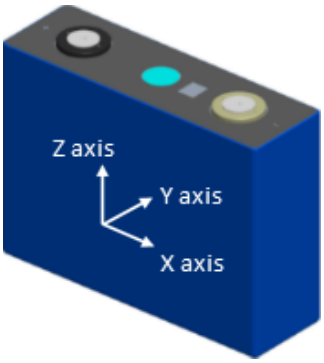


Fig. 6 Diagram of LF100LA cell directions

图 6 LF100LA 电池方向示意图

4.9 After charging, the cell should be used as soon as possible to avoid loss of usable capacity due to self-discharge. If long-term storage is required, adjust the cell SOC to 30% ~ 40%. The recommended storage conditions are: 0℃ ~ 25℃, relative humidity ≤ 60%.

电池进行充电后，需尽快使用，以免因自放电而造成可用容量损失。若需要存储，应将 SOC 调整为 30% ~ 40% SOC。推荐的存储条件为：0℃ ~ 25℃，相对湿度≤ 60%。

4.10 The state of charge (SOC, capacity state) of the cell should be kept at 15% ~ 40% during storage. In order to prevent the performance differences after long-term storage (more than three months), perform a standard charge-discharge cycle every 3 months. It is recommended that the storage time after receiving the cells should not exceed half a year to avoid quality problems due to storage overdue.

电池单体贮存时，荷电态（SOC，容量状态）应保持在 15%～40%，长期贮存（三个月以上）时为防止电芯性能差异，应每 3 个月进行一次标准充放电循环；建议收到货后储存时间不超过半年，避免存储超期造成的质量问题。

4.11 It is not allowed to ship with inflammable, explosive and corrosive articles in the same vehicle during transportation, and stacking is prohibited during large package transportation; The product shall not be exposed to rain, snow and other liquid substances without any protection, or suffer from mechanical damage;

在运输过程中不允许和易燃、易爆、易腐蚀的物品同车装运，大包装运输过程中禁止堆叠；产品不允许经受雨、雪或液体物质的淋袭与机械损伤；

4.12 While handling, lift trucks or special tools shall be used to load and unload products; Handle with care, and do not throw or squeeze, which may cause cell damage or personal injury. It is strictly prohibited to put cells together with corrosive substances such as acid and alkali.

产品装卸时，须采用升降车或专用工具对产品进行上下车；要轻取轻放，不得扔掷、挤压，造成电池损坏或对人身的意外伤害，严禁与酸碱等腐蚀物品放在一起。

5 Safety Precautions 安全防范

- It is strictly forbidden to immerse the cell in water. When it is not in use, it should be placed in a cool and dry environment.

  严禁将电池浸入水中，保存不用时，应放置于阴凉干燥的环境中。

- Do not over-charge the cell. Otherwise, cell overheating and fire may occur. During cell installation and use, hardware and software must be protected against multiple over-charge failures. See 4.3 of this specification for the minimum requirements of protection.

  禁止电池过充，否则，可能引起电池过热和火灾事故的发生。在电池安装和使用中，硬件和软件需实行多重过充失效安全保护。最低保护要求见本规格书 4.3 条。

- It's necessary to set a reasonable charging time limit, otherwise, the cell may overheat, resulting in thermal runaway or fire. BMS management failure of this kind shall be considered during module design.

  应对电池持续充电时长进行合理限制。充电时间过长电池可能会出现过热现象，进而引起热失控和火灾。在模组设计时需考虑此类 BMS 管理失效情况。

- If improper charge termination occurs, the root reasons shall be found and resolved before reuse to avoid negative effects on electrical and safety performances.

  电池非正常终止充电时，为避免未知原因对电池性能及安全的影响，需明确根本原因并彻底解决后再使用。

- It is forbidden to over-discharge. During the normal use of the cell, charge the cell regularly to keep the voltage above 2.8 V, so as to avoid over-discharge.

  禁止过放电。在电池正常使用过程中，为防止过放电，电池应定期充电，将电压维持在 2.8 V 以上。

- It is forbidden to use or place the cell at a high temperature environment. Otherwise, cell overheat, function failure or life shorten may occur.

  禁止在高温下使用或放置电池，否则可能会引起电池过热或功能失效、寿命减短。

- The temperature of the cell shall not exceed 65℃ in any normal use, otherwise the BMS must shut down the cell and stop cell operation.

  在任何正常的使用情况下，电池温度不能超过 65℃，如果电池温度超过 65℃，电池管理系统需关闭电池，停止电池运行。

- Please use a special charger for lithium-ion batteries when charging.

  充电时请选用锂离子电池专用充电器。

- During use, please connect the positive and the negative of the cell strictly according to the labels and instructions, and reverse charging is forbidden.

  在使用过程中，严格按照标示和说明连接电池正负极，禁止反向充电。

- It is forbidden to use metal to directly connect the positive and the negative of the cell to short-circuit. Otherwise, strong current and high temperature may cause personal injury or fire.

  禁止用金属直接连接电池正负极短路，否则强电流和高温可能导致人身伤害或者火灾。

- It is forbidden to transport or store the cell with metal, such as hairpins, necklaces, etc.

  禁止将电池与金属，如发夹、项链等一起运输或贮存。

- It is forbidden to knock, throw, step on or bend the cell.

  禁止敲击或抛掷、踩踏和弯折电池等。

- It is forbidden to directly weld the cell.

  禁止直接焊接电池。

- It is forbidden to directly pierce the cell with nails or other sharp objects.

  禁止用钉子或其它利器刺穿电池。

- Prevent the cell from mechanical shock, collision and pressure impact. Otherwise, the cell may be short-circuited internally, resulting in high temperature and fire.

  尽力保护电池，使其免受机械震动、碰撞及压力冲击，否则电池内部可能短路，产生高温和火灾。

- It is forbidden to use it in places with strong static electricity and strong magnetic fields; otherwise cell safety protection devices may be damaged and cause safety hazards.

  禁止在强静电和强磁场的地方使用，否则易破坏电池安全保护装置，带来安全隐患。

- Customer shall securely fix the cell to a solid surface and bind the power cord in a proper place to avoid arcing and sparks caused by friction.

客户应将电池安全地固定在固体平面上，并将电源线安全地束缚在合适的位置，以避免摩擦而引起电弧和火花。

- It is forbidden to use plastic for cell encapsulation and electrical connection. Improper electrical connection may cause overheat during cell use.

严禁用塑料封装电池或用塑料进行电气连接。不正确的电气连接方式可能会造成电池使用过程中发生过热现象。

- If the cell leaks and the electrolyte spills onto the skin or clothes, immediately wash the affected area with running water. If the cell leaks and the electrolyte enters the eyes, mouth, nose and other open parts of the human body, immediately wash with plenty of water and seek medical treatment immediately, otherwise serious injuries will be caused to the human body. No person or animal is allowed to swallow any part of the cell or any substance contained in the cell.

如果电池漏液，电解液溅到皮肤或衣服上，应立即用流动的水清洗受影响区域，如果电池发生泄露，电解液进入眼睛、口、鼻等人体开放部位，应立即用大量清水冲洗，并马上送医治疗，否则会对人体造成严重伤害。禁止任何人或动物吞食电池的任何部位或电池所含物质。

- If the cell emits peculiar smell, heat, discoloration, deformation, or any abnormality during use, storage, or charging, immediately remove the cell from the device or charger and stop using it.

如果电池发出异味、发热、变色、变形或使用、贮存、充电过程中出现任何异常，立即将电池从装置或充电器中移离并停用。

- It is prohibited to disassemble the product without the written consent of EVE.

  未经 EVE 书面同意，禁止私自拆解产品。

6 Disclaimer 免责声明

- If the product demand party or user does not use the product in accordance with the provisions of this specification, EVE will no longer take any relevant responsibility such as product quality assurance liability and loss compensation caused thereby. In case of any negative impact on EVE's reputation due to the above-mentioned actions, EVE reserves the right to investigate the legal liability of the product demand party.

  如果由于产品需求单位或使用者不按本说明书中的规定进行使用，EVE 不再承担产品质量保证责任及由此引起的损失赔偿等一切相关责任。因前述行为，对 EVE 的声誉造成负面影响的，EVE 保留追究产品需求单位法律责任权利。

- EVE reserves the right to modify the product specification. Before ordering EVE products, customers need to confirm the latest status of the products in advance with EVE.

  EVE 保留对本产品规格书修改的权利。客户在订购 EVE 产品前，需与 EVE 提前确认产品的最新状态。

- This specification is written in both Chinese and English. In case of any difference between the Chinese version and the English version, the Chinese version shall prevail.

Model 型号	LF100LA	Specification No. 规格书编号	PBRI-LF100L-V2.1-D06-01	Version 版本	C

本规格书以中英双语书就，如中英文约定存在冲突时，以中文版本为准。

7 Risk Warning 风险警告

7.1 Warning Declaration 警示声明

Warning 警告

The cell has potential hazards. Please take proper precautions when operating and maintaining the cell!

The cell must be operated with proper tools and protective equipment.

Cell maintenance must be performed by professionals with cell expertise and safety training.

Failure to comply with these warnings may result in multiple disasters.

电池存在潜在的危险，在操作和维护时必须采取适当的防护措施！

必须使用正确的工具和防护装备操作电池。

电池的维护必须由具有电池专业知识并经过安全培训的人士执行。

不遵守上述警告可能造成多种灾难。

7.2 Types of Hazards 危险类型

Customer must be aware of the following potential hazards in the use and operation of cells:

客户知悉在电池使用和操作过程中存在以下潜在的危险：

a) There is a risk from electric shocks or electric arcs during operation.

存在受到电击或者电弧伤害的风险。

b) There is a risk from the electrolyte or other chemicals.

存在受到电解液或其他化学品危害的风险。

c) Proper operation methods and protective equipment shall be selected to avoid accidental short circuit, leading to electric arcs, explosion or thermal runaway.

为防止发生意外短路，造成电弧、爆炸或热失控，需选择合适的操作方法及防护装备。

8 Others 其他

8.1 Rights and Obligations of EVE EVE 的权利和义务

- EVE shall inspect according to the inspection standards in the protocol signed with the customer, and the products provided shall meet the requirements of various parameters in the protocol;

  EVE 按照与客户签订的协议中的检验标准进行检验，提供产品应符合协议中各项参数要求；

- EVE shall provide customer with stable and reliable products confirmed by both parties;

  EVE 向客户提供双方确认的、稳定可靠的产品；

➢ EVE is obliged to provide high-quality services for its products, and the service standards shall be in accordance with the standards promised by EVE;

EVE 有义务为其产品提供优质的服务，服务标准按 EVE 所承诺的标准服务；

➢ EVE shall provide timely technical support and service in case of any problem or failure during the use and maintenance of system products by customer.

在客户对系统产品的使用及维护过程中遇到问题或故障时，EVE 进行及时的技术支持及服务。

8.2 Rights and Obligations of Customer 客户的权利和义务

➢ Customer must conduct production in strict accordance with the technical data provided by EVE, and strictly implement the cell prevention measures, safety limits and cell operation instructions in the technical data provided by EVE.

客户应严格按照 EVE 提供的技术资料进行生产，严格执行 EVE 所提供的技术资料中的电池防范措施、安全限制和电池操作说明。

➢ Customer has the obligation to ensure the safety of products by EVE and shall take corresponding fire prevention, waterproof and other measures.

客户有义务保证 EVE 产品的安全，应采取相应的防火、防水等措施。

➢ Customer has the obligation to make fair and detailed use records and monitoring data of product operation for EVE's products, which can be used as a reference for the division of product quality responsibilities. If there is no complete monitoring data within the service life of the battery system, EVE shall not be responsible for product quality assurance.

客户有义务对 EVE 的产品做出公平、公正详细的使用记录及产品运转的监测数据，用作于产品质量责任划分的参考，不具备完整的电池系统使用期限内的监测数据的，EVE 不承担产品质量保证责任。

➢ Customer has the obligation to notify EVE's personnel to be present and be informed of the actual situation when products of EVE operate abnormally.

客户有义务在 EVE 产品运作异常时，在可能的情况下通知 EVE 人员到场，了解实际情况。

➢ In the process of product manufacturing, customer shall take all corresponding responsibilities for the problems or accidents caused by the operation in violation of the safety rules, the use beyond the conditions specified in this technical agreement, or the combination of the product and the circuit (not the quality defects of the product itself).

客户在产品制造过程中，因违反安全守则操作、在本规格书说明条件之外使用及产品与电路等搭配（非产品自身质量缺陷）所产生的问题或事故，其责任与 EVE 无关，应由客户承担相应的责任。

➢ Any matter not mentioned in this specification must be negotiated and determined by both parties.

任何本规格书中未提及的事项，须经双方协商确定。

8.3 Confidentiality agreement 保密协议

The customer shall keep the cooperation content highly confidential. Without the permission of EVE, the customer shall not disclose any content of the technical agreement to a third party. Otherwise, the customer will be held responsible according to relevant laws.

客户应对合作内容高度保密，未经 EVE 许可，不得向第三方透露规格书的任何内容，否则，将依照相关法律追究责任。

9 Contact Information 联系方式

Address: EVE Power Co., Ltd., No. 68 Jingnan Avenue High-Tech Zone, Duodao District, Jingmen, Hubei.

Tel：86-0724-6079688

Website：http://www.evepower.com

联系地址：湖北省荆门市经济开发区高新区 掇刀区荆南大道 68 号，湖北亿纬动力有限公司

联系电话：86-0724-6079688

网址：http://www.evepower.com

Model 型号	LF100LA	Specification No. 规格书编号	PBRI-LF100L-V2.1-D06-01	Version 版本	C

Appendix 1: Cell Drawing of LF100LA 附录 1：LF100LA 电池图纸

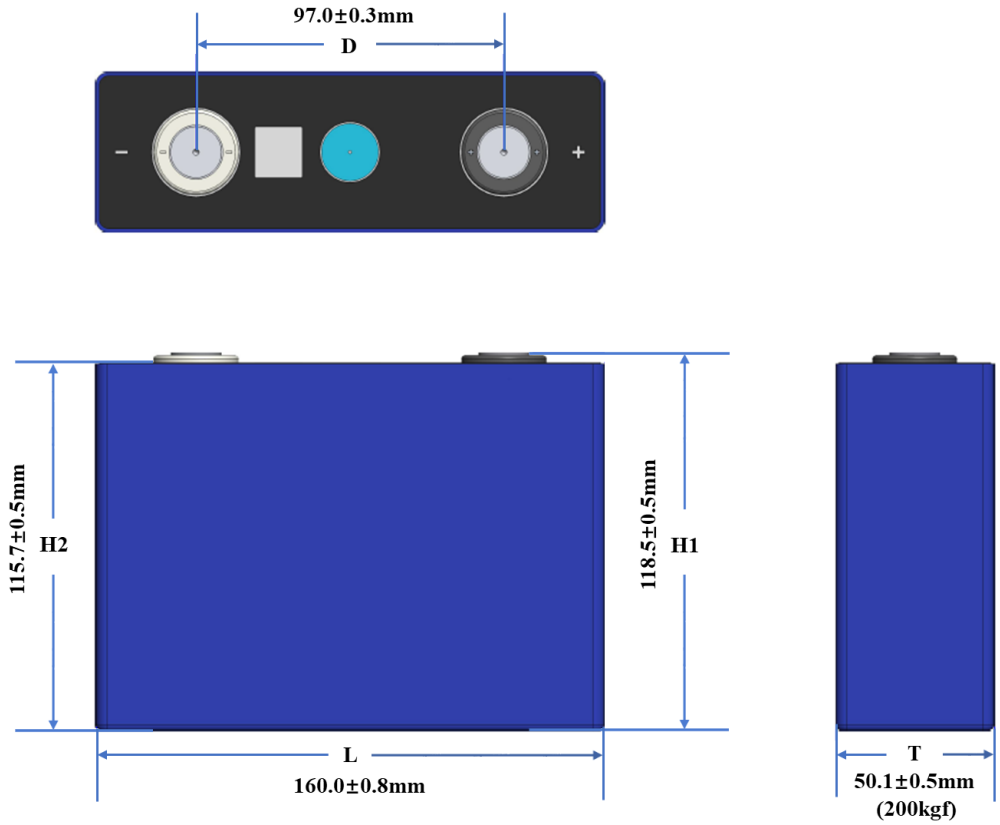


Fig. 7 Diagram of cell size and appearance

图 7 电池尺寸及外观图

Appendix 2: Test Conditions and Methods 附录 2：试验条件及方法

1.1 Environmental Conditions 环境条件

Unless otherwise specified, the test should be carried out in an environmental temperature of 25℃ ±2℃, relative humidity of 10% ~ 90%, and atmospheric pressure of 86 kPa to 106 kPa. The room temperature mentioned in this specification refers to 25℃ ± 2℃.

除另有规定外，试验应在温度为 25℃ ±2℃，相对湿度 10% ~ 90%，大气压力为 86 kPa ~ 106 kPa 的环境中进行。本规格书所提到的室温，是指 25℃ ±2℃。

1.2 Measuring Instruments 测量设备

The accuracy of measuring instruments shall meet the following requirements:

试验装置应符合下列要求：

A. Voltage measuring device 电压测量装置：±0.05 % FS；

B. Current measuring device 电流测量装置：±0.05 % FS；

C. Temperature measuring device 温度测量装置：± 1℃；

D. Dimension measuring device 尺寸测量装置：±0.01 mm；

E. Weight measuring device 重量测量装置：±0.1 g。

Note: During the testing process, voltage, current, ambient temperature, and cell temperature must be recorded. It is recommended to collect the cell temperature at the terminal and the center of the large surface (or side surface).

备注：测试过程中，均需记录电压、电流、环境温度和电池温度，电池温度采集点建议布置在极柱和大面（或侧面）中心处。

1.3 Test Clamp Preparation and Installation 测试夹具准备和安装

1.3.1 Ordinary Steel Clamp 普通钢夹具

The single cell shall be clamped with steel splints (thickness: ≥8 mm). The splints need to cover the large surfaces of the cell and be fixed with 4 M8 bolts. All sides of the splints need to be covered with insulating film. Clamp as shown below:

单体电池需采用钢夹板（厚度：≥8 mm）固定，夹板需要覆盖住电池大面，夹板之间采用 4 个 M8 螺栓固定，夹板各个面需要有绝缘膜，夹具工装如下图所示：

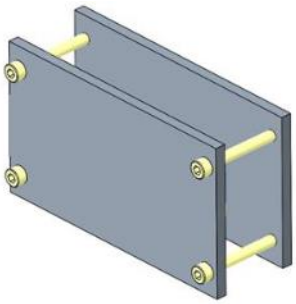


Fig. 1 Diagram of the clamp
图 1 夹具示意图



Fig. 2 Diagram of the clamp covered with insulation film
图 2 电池夹具包绝缘膜图

Place the cell (15% ~ 40% SOC) which is covered with blue film (material: PET, thickness: 0.1 mm) and top film (material: PC, thickness: 0.3 mm) in the middle of the splint, fix the clamp with 4 M8 bolts to ensure the initial compression force on the cell is 200 kgf ±20 kgf,. Then use a micrometer to measure the thickness of the cell (at upper, middle and bottom) with splints, and the thickness tolerance shall be controlled within 0.3 mm.

将包覆有蓝膜（材质：PET，厚度 0.1 mm）和顶部贴片（材质：PC，厚度 0.3 mm）的电池（15% ~ 40% SOC）准备好后，置于夹具中间，使用 4 个 M8 螺栓固定钢夹具保证电池受夹具初始预紧力为 200 kgf ±20 kgf，需采用千分尺测量带夹板电池的厚度（上、中、下），厚度差异需控制小于等于 0.3 mm。

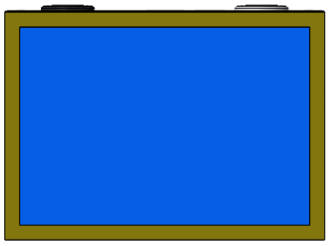


Fig. 3 Diagram of cell filming
图 3 电池包膜示意图



Fig. 4 Diagram of cell with clamp
图 4 电池带夹具示意图

1.3.2 Swelling Force Clamp 膨胀力夹具

Place the cell (15% ~ 40% SOC) which is covered with blue film (material: PET, thickness: 0.1 mm) and top film (material: PC, thickness: 0.3 mm) in the middle of the splint, and adjust the swelling force clamp pre-tensioning device until the initial compression force is (200 kgf ±20 kgf). Then use a micrometer to measure the thickness of the cell (at upper, middle and bottom) with splints, and the thickness tolerance shall be controlled within 0.3 mm.

将包覆有蓝膜（材质：PET，厚度 0.1 mm）和顶部贴片（材质：PC，厚度 0.3 mm）的电池（50% SOC）准备好后，置于膨胀力夹具中间，调节膨胀力夹具预紧装置，使传感显示装置压缩力为 200 kgf ±20 kgf，上夹板后，需采用千分尺测量带夹板电池的厚度（上、中、下），厚度差异需控制小于等于 0.3 mm。

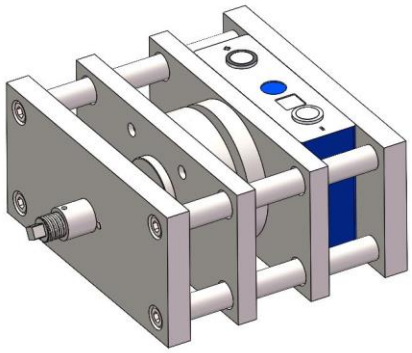


Fig. 5 Diagram of cell with swelling force clamp
图 5 电池带膨胀力夹具示意图

1.4 Dimensions 尺寸

Test instrument: Automatic wrapping machine;

试验设备：自动包膜机；

Test method: Use the wrapping machine to measure the length, width and height of the cell. And apply a 200 kgf ±20 kgf force on it.

试验方法：使用自动包膜机测试电池长度、宽度和高度，施加 200 kgf ±20 kgf 的压力。

The thickness of the cell will increase as the SOC increases as well as with the using time. The thickness here indicates the thickness of the cell at delivery.

电池厚度随着 SOC 增加会有所增加，随着使用时间增加会有所增加，此处厚度指出货时电池的厚度。

1.5 Weight 重量

Test instrument: Electronic scale;

试验设备：电子秤；

Test method: measure the weight of the cell with the electronic scale.

试验方法：使用电子秤测量电池的重量。

1.6 Internal Resistance 内阻

a. ACR: test the cell at delivery SOC with 1kHz sine wave current at room temperature.

a. 交流内阻（ACR）：在室温条件下，出货 SOC 电池采用 1 kHz 正弦波电流进行测试。

b. DCR: 1) The battery is discharged to 2.5V by standard discharge mode (1.8), 2) The battery is charged to 3.65V by standard charge mode (1.7). 3) Then discharge at a constant current of 0.2C for 150 minutes (adjust the SOC to 50%) and rest for 30 minutes. 4) The battery is discharged at a constant current of 0.2C for 30s, then further discharged with a constant current of 1C for 5s. There is no interval between the two constant current discharges at different rates. Record the voltages at the end of 0.2C and 1C discharges as $V_{0.2C}$ and V_{1C} respectively. 5) The DCR.is calculated by

$(V_{0.2c}-V_{1c})/(I_{1c}-I_{0.2c})$.

b. 直流内阻（DCR）：1）电池按照（1.8）进行标准放电；2）电池按照（1.7）进行标准充电；3）电池以 0.2C 放 2.5h（调整 SOC 为 50%），搁置 30min；4）电池以 0.2C 恒流放电 30 秒，接着以 1C 恒流放电 5 秒，两个倍率恒流放电之间无搁置， 分别记录 0.2C 与 1C 放电结束时电压为 $V_{0.2C}$ 与 V_{1C}，5）0.2C 放电与 1C 放电结束时的电压差除以对应电流差值的比（$V_{0.2c}-V_{1c}$)/($I_{1c}-I_{0.2c}$）即为该 SOC 下的 DCR。

1.7 Standard Charge 标准充电方式

The battery is charged at a constant current of 0.2C to 3.65V under the condition of an ambient temperature of 25℃±2℃, and then transfers to constant voltage charging at 3.65V until the charging current is less than or equal to 0.05C, and rest for 30min.

标准充电是在环境温度 25℃±2℃的条件下，对电池以 0.2C 的电流恒流充电至 3.65V，然后在 3.65V 下转恒压充电，直至充电电流小于等于 0.05C，搁置 30min。

1.8 Standard Discharge 标准放电方式

The battery is discharged at 0.2C constant current until the voltage reaches 2.5V cutoff under an ambient temperature of 25 ℃ ±2 ℃, and rest for 30min.

标准放电是在环境温度 25℃±2℃的条件下，电池以 0.2C 的电流恒流放电，放电至电压达到 2.5V 截止，搁置 30min。

1.9 0.2C Discharge Capacity 0.2C 放电容量

Under the condition of an ambient temperature of 25℃±2℃, 1) the battery is discharged at a constant current of 0.2C to 2.5V and rests for 30min, 2) then charged to 3.65V by standard charge mode (1.7) and then discharged to 2.5V a current of 0.2C, the final discharge capacity C_0 are 0.2C capacity. When the capacity of the first lap does not meet the specification, it is allowed to retest for 3 times.

在环境温度 25±2℃的条件下，1）电池以 0.2C 恒流放电至 2.5V，静置 30min；2）电池按照（1.7）进行标准充电；3）再以 0.2C 恒流放电至截止电压 2.5V，记录 0.2C 容量为 C_{0}。当首圈容量不满足规格，允许复测 3 次。

1.10 High/Low Temperature Discharge Performance 高/低温放电性能

1.10.1 -20℃ Discharge Performance -20℃放电性能

1) The battery is discharged to 2.5V by standard discharge mode (1.8), 2) The battery is charged to 3.65V by standard charge mode (1.7), 3) The battery stands at -20 ℃ for 4h and then discharged at a constant current of 0.5C to 2.0V, rests for 30min, record the discharge capacity C_5, C_5/C_1 is the capacity retention rate.

1）电池按照（1.8）进行标准放电；2）电池按照（1.7）进行标准充电；3）电池在-20℃环境下搁置 4 小时，然后以 0.5C 电流恒流放电至 2.0V，静置 30min，记录放电容量 C_5，C_5/ C_1（25℃ 0.5C 放电容量）即为容量保持率。

1.10.2 0℃ Discharge Performance 0℃放电性能

1) The battery is discharged to 2.5V by standard discharge mode (1.8), 2) The battery is charged to 3.65V by standard charge mode (1.7), 3) The battery stands at 0 ℃ for 4h and then discharged at a constant current of 0.5C to 2.0V, rests for 30min, record the discharge capacity C_6, C_6/C_1 is the capacity retention rate.

1）电池按照（1.8）进行标准放电；2）电池按照（1.7）进行标准充电；3）电池在 0℃环境下搁置 4 小时，然后以 0.5C 的电流恒流放电至 2.0V，静置 30min，记录放电容量 C_6，C_6/C_1（25℃ 0.5C 放电容量）即为容量保持率。

1.10.3 10℃ Discharge Performance 10℃放电性能

1) The battery is discharged to 2.5V by standard discharge mode (1.8), 2) The battery is charged to 3.65V by standard charge mode (1.7), 3) The battery stands at 10 ℃ for 4h and then discharged at a constant current of 0.5C to 2.5V, rests for 30min, record the discharge capacity C_7, C_7/C_1 is the capacity retention rate.

1）电池按照（1.8）进行标准放电；2）电池按照（1.7）进行标准充电；3）电池在 10℃环境下搁置 4 小时，然后以 0.5C 的电流恒流放电至 2.5V，静置 30min，记录放电容量 C_7，C_7/C_1（25℃ 0.5C 放电容量）即为容量保持率。

1.10.4 55℃ Discharge Performance 55℃放电性能

1) The battery is discharged to 2.5V by standard discharge mode (1.8), 2) The battery is charged to 3.65V by standard charge mode (1.7), 3) The battery stands at 55 ℃ for 4h and then discharged at a constant current of 0.5C to 2.5V, rests for 30min, record the discharge capacity C_8, C_8/C_1 is the capacity retention rate.

1）电池按照（1.8）进行标准放电；2）电池按照（1.7）进行标准充电；3）电池在 55℃环境下搁置 4 小时，然后以 0.5C 的电流恒流放电至 2.5V，静置 30min，记录放电容量 C_8，C_8/C_1（25℃ 0.5C 放电容量）即为容量保持率。

1.11 Rate Discharge Performance 倍率性能

1) The battery is discharged to 2.5V by standard discharge mode (1.8), 2) The battery is charged to 3.65V by standard charge mode (1.7), 3) The battery stands at 25℃ for 4h and then discharged to 2.5V at a current of XC, the discharge capacity is recorded as C_Y, C_Y/C_2 is the capacity retention rate, and the battery stands still for Z minutes, then 4) repeats steps 2) to 3) to obtain the discharge capacity and capacity retention rate at different rate.

1）电池按照（1.8）进行标准放电；2）电池按照（1.7）进行标准充电；3）电池在 25℃下搁置 4h，再以 XC 恒流放电至截止电压 2.5V，其中放电容量记为 C_Y，C_Y/25℃ 1C 放电容量（C_2）即为容量保持率，搁置时间 Z；4）重复 2~3 工步，得到不同倍率的放电容量与容量保持率。

Note: For discharge rates X of 0.5C, 1C, 2C, and 2.5C, the corresponding recorded discharge capacity are C_1, C_2, C_3, and C_4 respectively。The corresponding standing time Z after discharge is 60 minutes, 120 minutes, 240 minutes, and 240 minutes respectively. (When X<0.5C, Z = 30 minutes; when 0.5C ≤X<1C, Z = 60 minutes; when X = 1C, Z = 120

minutes; when X>1C, Z = 240 minutes).

备注：倍率 X=0.5C、1C、2C 和 2.5C，其对应记录的放电容量分别为 C_1、C_2、C_3 和 C_4；其对应的放电后搁置 Z 分别为 60min、120min、240min、240min。（X<30min；0.5C≤X<1C 时，Z=60min；X=1C 时，Z=120min；X>1C 时，Z=240min）

1.12 Energy Efficiency 能量效率

The battery is discharged to 2.5V by standard discharge mode (1.8), 2) The battery is charged to 3.65V by standard charge mode (1.7), 3) The battery is discharged to 2.5V by standard discharge mode (1.8), 4) The battery is charged to 3.65V by standard charge mode (1.7), 5) The battery is discharged to 2.0V at a current of 1C at 25℃ and rests for 30min, 6) The battery is charged to 3.65V at a constant current of 0.5C, rests for 60min, record the charge energy E_0, 7) The battery is discharged to 2.0V at a constant current of 0.5C and rests for 60min, record the discharge energy E_1, energy efficiency $\eta=E_1/E_0$. Repeat steps from 6) to 7) for 4 times, Take the average value of η for 3 times after the cycle as the judgment basis.

1）电池按照（1.8）进行标准放电；2）电池按照（1.7）进行标准充电；3）电池按照（1.8）进行标准放电；4）电池按照（1.7）进行标准充电；5）电池在 25℃下以 1C 恒流放电至 2.0V，静置 30min；6）电池以 0.5 恒流充电至截止电压 3.65V，记录充电能量 E_0，搁置 60min；7）电池以 0.5C 恒流放电至 2.0V，记录放电能量为 E_1，搁置 60min；能量效率 $\eta=E_1/E_0$。循环步骤 6~7 四次，取循环后 3 次 η 平均值作为判定依据。

1.13 Floating Charge 浮充性能

1) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min; 2) The battery is charged to 3.65V at a constant current of 0.2C, and then switched to constant voltage charging at 3.65V, until the charging current is less than or equal to 0.05C, rests for 30min; 3）The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, record the discharge capacity as C_9; 4) The battery is charged to 3.4V at a constant current of 0.2C, and then switched to constant voltage charging at 3.4V, until the charging current is less than or equal to 0.05C, rests for 30min, 5) the battery is charged at a constant voltage of 3.4V at 25℃ for 15 years (The battery is charged and discharged by standard mode once every 15days), then the battery is discharged by standard discharge mode(1.8), record the discharge capacity as C_{10}, the capacity retention can be calculated as C_{10}/C_9.

1）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min；2）电池在 25℃下以 0.2C 恒流恒压电至 3.65V，截止电流 0.05C，静置 30min；3）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min；记录放电容量 C_9；4）电池以 0.2C 的电流恒流充电至 3.4V，然后在 3.4V 下转恒压充电，直至充电电流小于等于 0.05C，静置 30min；5）电池在 25℃下恒压充电至 3.4V 持续 15 年（每 15 天进行一次标准充放电），15 年后电池进行标准放电，记录放电容量为 C_{10}，容量保持率为 C_{10}/C_9。

1.14 Cycle life 循环寿命

1.14.1 25℃ 0.5C Cycle life 25℃ 0.5C 循环寿命

1) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, 2) The battery is placed in 25±2℃ for 4 hours, 3) The battery is charged to 3.65V at a constant current of 0.5C, and then switched to constant voltage charging at 3.65V, until the charging current is less than or equal to 0.05C, rests for 30min, 4) The battery is discharged to 2.5V at 0.5C constant current then rests for 30min, 5) Repeat steps from 3) to 4) until the discharge capacity of step 4) < 80% of the rated capacity, record the number of cycles.

1）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min；2）电池置于 25±2℃的环境中，搁置 4h；3）电池以 0.5C 的电流恒流充电至 3.65V，然后在 3.65V 下转恒压充电，直至充电电流小于等于 0.05C，搁置 30min；4）电池以 0.5C 恒流放电至 2.5V，搁置 30min；5）循环第 3~4 步，直至第 4 步放电容量< 80%额定容量为终止，记录充放电圈数。

1.14.2 25℃ 1C Cycle life 25℃ 1C 循环寿命

1) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, 2) The battery is placed in 25±2℃ for 4 hours, 3) The battery is charged to 3.65V at a constant current of 1.0C, and then switched to constant voltage charging at 3.65V, until the charging current is less than or equal to 0.05C, rests for 30min, 4) The battery is discharged to 2.5V at 1.0C constant current then rests for 30min, 5) Repeat steps from 3) to 4) until the discharge capacity of step 4) < 80% of the rated capacity, record the number of cycles.

1）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min；2）电池置于 25±2℃的环境中，搁置 4h；3）电池以 1.0C 的电流恒流充电至 3.65V，然后在 3.65V 下转恒压充电，直至充电电流小于等于 0.05C，搁置 30min；4）电池以 1.0C 恒流放电至 2.5V，搁置 30min；5）循环第 3~4 步，直至第 4 步放电容量< 80%额定容量为终止，记录充放电圈数。

1.14.3 45℃ 0.5C Cycle life 45℃ 0.5C 循环寿命

1) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, 2) The battery is placed in 45±2℃ for 4 hours, 3) The battery is charged to 3.65V at a constant current of 0.5C, and then switched to constant voltage charging at 3.65V, until the charging current is less than or equal to 0.05C, rests for 30min, 4) The battery is discharged to 2.5V at 0.5C constant current then rests for 30min, 5) Repeat steps from 3) to 4) until the discharge capacity of step 4) < 80% of the rated capacity, record the number of cycles.

1）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min；2）电池置于 45±2℃的环境中，搁置 4h；3）电池以 0.5C 的电流恒流充电至 3.65V，然后在 3.65V 下转恒压充电，直至充电电流小于等于 0.05C，搁置 30min；4）电池以 0.5C 恒流放电至 2.5V，搁置 30min；5）循环第 3~4 步，直至第 4 步放电容量< 80%额定容量为终止，记录充放电圈数。

1.14.4 45℃ 1C Cycle life 45℃ 1C 循环寿命

1) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, 2) The battery is placed in 45±2℃ for 4 hours, 3) The battery is charged to 3.65V at a constant current of 1.0C, and then switched to constant voltage charging at 3.65V, until the charging current is less than or equal to 0.05C, rests for 30min, 4) The battery is discharged to 2.5V at 1.0C constant current then rests for 30min, 5) Repeat steps from 3) to 4) until the discharge capacity of step 4) < 80% of the rated capacity, record the number of cycles.

1）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min；2）电池置于 45±2℃的环境中，搁置 4h；3）电池以 1.0C 的电流恒流充电至 3.65V，然后在 3.65V 下转恒压充电，直至充电电流小于等于 0.05C，搁置 30min；4）电池以 1.0C 恒流放电至 2.5V，搁置 30min；5）循环第 3~4 步，直至第 4 步放电容量< 80%额定容量为终止，记录充放电圈数。

1.15 Capacity Retention and Recovery 荷电保持与恢复

1.15.1 25℃ Capacity Retention and Recovery 25℃荷电保持与恢复

1) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, 2) The battery is charged to 3.65V at a constant current of 0.2C, and then switched to constant voltage charging at 3.65V, until the charging current is less than or equal to 0.05C, rests for 30min, 3) The battery is discharged at a current of 0.2C to 2.5V at 25℃ and the discharged capacity C_{11} is recorded; then rests for 30min, 4) The battery is charged to 3.65V at a constant current of 0.2C, and then switched to constant voltage charging at 3.65V, until the charging current is less than or equal to 0.05C, rests for 30min, 5) The batteries are stored in the temperatures of 25℃ for 28 days. 6) The battery is put aside for 4 hours at an ambient temperature of 25℃, 7) and then the battery is discharged to 2.5V at a current of 0.2C and rests for 30min, record the discharge capacity as C_{12}, 8) Repeat the step 2), 9) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, record the discharge capacity as C_{13}, Capacity retention rate=C_{12}/C_{11}×100%, capacity recovery rate=C_{13}/C_{11}×100%.

1) 电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min；2) 电池在 25℃下以 0.2C 恒流恒压充电至 3.65V/0.05C 截止，静置 30min；3）电池在 25℃下以 0.2C 恒流放电至 2.5V，记录放电容量 C_{11}；静置 30min；4）电池在 25℃下以 0.2C 恒流恒压充电至 3.65V/0.05C 截止，静置 30min；5）电池在 25℃环境下存储 28d；6）电池存储后取出后在 25℃下搁置 4 小时；7）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min，取放电容量 C_{12}；8）重复步骤 2；9）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min，记录容量 C_{13}，容量保持率=C_{12}/C_{11}，容量恢复率=C_{13}/C_{11}。

1.15.2 45℃ Capacity Retention and Recovery 45℃荷电保持与恢复

1) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, 2) The battery is charged to 3.65V at a constant current of 0.2C, and then switched to constant voltage charging at 3.65V, until the charging current is less than or equal to 0.05C, rests for 30min, 3) The battery is discharged at a current of 0.2C to 2.5V at 25℃ and the discharged

capacity C_{14} is recorded; then rests for 30min, 4) The battery is charged to 3.65V at a constant current of 0.2C, and then switched to constant voltage charging at 3.65V, until the charging current is less than or equal to 0.05C, rests for 30min; 5) The batteries are stored in the temperatures of 45℃ for 28 days. 6) The battery is put aside for 4 hours at an ambient temperature of 25℃, 7) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, record the discharge capacity as C_{15}, 8) Repeat the step 2), 9) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, record the discharge capacity as C_{16}, Capacity retention rate=C_{15}/C_{14}×100%, capacity recovery rate=C_{16}/C_{14}×100%.

1）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min；2）电池在 25℃下以 0.2C 恒流恒压充电至 3.65V/0.05C 截止，静置 30min；3）电池在 25℃下以 0.2C 恒流放电至 2.5V，记录放电容量 C_{14}；静置 30min； 4）电池在 25℃下以 0.2C 恒流恒压充电至 3.65V/0.05C 截止，静置 30min；5）电池在 45℃环境下存储 28d；6）电池存储后取出后在 25℃下搁置 4 小时；7）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min，取放电容量 C_{15}；8）重复步骤 2；9）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min，记录容量 C_{16}，容量保持率=C_{15}/C_{14}，容量恢复率=C_{16}/C_{14}。

1.16 Storage 存储性能

1.16.1 25℃ Storage 25℃存储

1) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, 2) The battery is charged to 3.65V at a constant current of 0.2C, and then switched to constant voltage charging at 3.65V, until the charging current is less than or equal to 0.05C, rests for 30min, 3) The battery is discharged at a current of 0.2C to 2.5V at 25℃ and rests for 30min, Record the discharge capacity C_{17} as the initial capacity. 4) The battery is charged to 3.65V at a constant current of 0.2C, and then switched to constant voltage charging at 3.65V, until the charging current is less than or equal to 0.05C, rests for 30min; 5) Discharge at 0.2C to 50% of C_{17}, 6) The batteries are stored in the temperatures of 25℃ for 28 days, 7) The battery is put aside for 4 hours at an ambient temperature of 25℃, 8) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, 9) Repeat the step 4), 10) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, record the discharge capacity as C_{18}, Capacity recovery rate=C_{18}/C_{17}.

1）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min；2）电池在 25℃下以 0.2C 恒流恒压充电至 3.65V/0.05C 截止，静置 30min；3）电池在 25℃下以 0.2C 恒流放电至 2.5V；静置 30min；记录放电容量 C_{17} 作为初始容量。4）电池在 25℃下以 0.2C 恒流恒压充电至 3.65V/0.05C 截止，静置 30min；5）0.2C 放电至 50% C_{17}，6）电池在 25℃环境下存储 28d；7）电池存储后取出后在 25℃下搁置 4 小时；8）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min；9）重复步骤 4)；10）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min，记录容量 C_{18}，容量恢复率=C_{18}/C_{17}。

1.16.2 45℃ Storage 45℃存储

1) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, 2) The battery is charged to 3.65V at a constant current of 0.2C, and then switched to constant voltage charging at 3.65V, until the charging current is less than

or equal to 0.05C, rests for 30min, 3) The battery is discharged at a current of 0.2C to 2.5V at 25℃ and rests for 30min, Record the discharge capacity C_{19} as the initial capacity. 4) The battery is charged to 3.65V at a constant current of 0.2C, and then switched to constant voltage charging at 3.65V, until the charging current is less than or equal to 0.05C, rests for 30min; 5) Discharge at 0.2C to 50% of C_{19}, 6) The batteries are stored in the temperatures of 45℃ for 28 days, 7) The battery is put aside for 4 hours at an ambient temperature of 25℃, 8) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, 9) Repeat the step 4), 10) The battery is discharged to 2.5V at a current of 0.2C and rests for 30min, record the discharge capacity as C_{20}, Capacity recovery rate=C_{20}/C_{19}.

1）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min；2）电池在 25℃下以 0.2C 恒流恒压充电至 3.65V/0.05C 截止，静置 30min；3）电池在 25℃下以 0.2C 恒流放电至 2.5V；静置 30min；记录放电容量 C_{19} 作为初始容量。4）电池在 25℃下以 0.2C 恒流恒压充电至 3.65V/0.05C 截止，静置 30min；5）0.2C 放电至 50% C_{19}，6）电池在 45℃环境下存储 28d；7）电池存储后取出后在 25℃下搁置 4 小时；8）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min；9）重复步骤 4)；10）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min，记录容量 C_{20}，容量恢复率=C_{20}/C_{19}。

1.17 Swelling Force Test 膨胀力测试

Prepare and install the clamp according to (1.3.2) before the test.

测试前按照（1.3.2）进行夹具准备和安装。

Cycle according to (1.14.4), and record cell swelling force and discharge capacity retention rate during test process.

按照（1.14.4）进行循环，记录每圈充放电过程中的电池最大膨胀力和放电容量保持率。

1.18 Cell temperature rise without wind 无风温升

1) Discharge the battery at a constant current of 0.2C to 2.5V at 25 ℃, and let it stand for 30 minutes. 2) The battery is charged to 3.65V at a constant current of 0.2C, and then switched to constant voltage charging at 3.65V, until the charging current is less than or equal to 0.05C, rests for 30min. 3) The battery is discharged at a current of 0.2C to 2.5V at 25℃ and rests for 30min. 4) Set the temperature chamber to 35 ℃ and then pause the test. Cover the battery cell with a wind shield and let it stand for 4 hours. 5) The battery is charged to 3.65V at a constant current of 0.2C, and then switched to constant voltage charging at 3.65V, until the charging current is less than or equal to 0.05C, rests for 2 hours. 6) Discharge the battery at a constant current of 1C to 2.5V at 35 ℃. Record the temperature rise during the discharge process (on the large surface of the battery cell) and let it stand for 2 hours.

1）电池在 25℃下以 0.2C 恒流放电至 2.5V，静置 30min；2）电池以 0.2C 恒流恒压充电至 3.65V/0.05C 截止，搁置 30min；3）电池在 25℃下以 0.2C 恒流放电至 2.5V；4）温箱配置为 35℃，暂停测试，用防风罩罩住电芯搁置 4h；5）电池在 35℃下 1C 恒流恒压充电至 3.65V/0.05C 截止，搁置 2h；6）电池在 35℃下 1C 恒流放电至 2.5V，记录放电过程温升（电芯大面），搁置 2h。

1.19 Safety Performance 安全性能

1.19.1 Over-charge Test 过充电试验

Under the condition of ambient temperature of 25 ℃ ±2 ℃, discharge the battery cell completely. Use the maximum continuous charging current (stated in the product specification) to charge it to 1.2 times the maximum sustainable charging voltage. When the temperature drops to the ambient temperature or the temperature change of the sample is less than 10 ℃ within 30 minutes, stop the test.

在环境温度 25℃±2℃的条件下，放空电电芯，使用最大持续充电电流（产品规格书中说明），充电至 1.2 倍的最大可持续充电电压，温度下降至环境温度或者样品在 30 min 内温度变化小于 10℃时，停止测试。

(Refer to IEC62619-2022 Alkaline and non-acidic electrolyte batteries - Safety requirements for industrial secondary lithium-ion batteries).

（参考 IEC62619-2022《碱性和非酸性电解液电池-工业用二次锂离子电池安全要求》）。

1.19.2 Over-discharge Test 过放电试验

Under the condition of ambient temperature of 25 ℃ ±2 ℃, discharge the fully depleted battery cell and force it to discharge at a constant current of 1C for 90 minutes. Then observe for 1 hour.

在环境温度 25℃±2℃的条件下，放空电电芯，以 1C 恒定电流强制放电 90 min，观察 1h。

(Refer to IEC62619-2022 Alkaline and non-acidic electrolyte batteries - Safety requirements for industrial secondary lithium-ion batteries).

（参考 IEC62619-2022《碱性和非酸性电解液电池-工业用二次锂离子电池安全要求》）。

1.19.3 External Short-circuit Test 外短路试验

Under the condition of ambient temperature of 25 ℃ ±2 ℃, the battery is charged in accordance with (1.7). Connect the positive and negative poles with an external resistance of 30 ± 10mΩ and short-circuit. The battery is kept in short-circuit condition for 6 hours or until the temperature of the casing drops to 80% of the maximum temperature. The test can be stopped if one of the above two conditions is satisfied.

在环境温度 25℃±2℃的条件下，电池按照（1.7）进行标准充电，使用 30±10 mΩ的外部电阻连接正极和负极极柱，进行短路，电池在短路的工况下保持 6 个小时，或直至壳体的温度下降至最大温度的 80%，上述两个条件满足一种就可以停止测试。

(Refer to IEC62619-2022 Alkaline and non-acidic electrolyte batteries - Safety requirements for industrial secondary lithium-ion batteries).

（参考 IEC62619-2022《碱性和非酸性电解液电池-工业用二次锂离子电池安全要求》）。

1.19.4 Crush Test 挤压试验

1) The battery is charged in accordance with (1.7).

1) 电池按照（1.7）进行标准充电；

2) The battery cell is left idle in an environment of (20±5) ℃ until the casing reaches the ambient temperature.

2) 电芯在 20℃±5℃环境中搁置至外壳达到环境温度；

3) The battery cell is placed between two planes and squeezed. The squeezing force is applied by a hydraulic plunger or a similar force structure. Once the force reaches (13±1) kN, the force is released.

3) 电芯放在两个平面之间进行挤压，挤压力由液压柱塞或类似的力结构施加，一旦力达到（13±1）kN，力就释放。

(Refer to UL1642-2022 Lithium battery safety standards).

（参考 UL1642-2022《锂电池安全标准》）。

1.19.5 Drop Test 跌落试验

Under the condition of ambient temperature of 25℃ ±2℃, the battery is discharged in accordance with (1.8), and then charged in accordance with (1.7). The battery cell is dropped from a height of 1 meter onto a flat concrete or metal floor in a random direction (for metal floors, appropriate measures should be taken to avoid external short circuit between the battery cell and the floor) three times. Observe for 1 hour.

在环境温度 25℃±2℃的条件下，电池按照（1.8）进行标准放电，再按照（1.7）进行标准充电，电芯从 1 m 的高度按照随机方向跌落到平坦的混凝土或金属地板上（金属地板需采用适当的措施避免电芯与地板发生外部短路）三次；观察 1h。

(Refer to UL1973-2022 Battery safety standards for stationary energy storage equipment and power-assisted power supply applications).

（参考 UL1973-2022《用于固定储能设备和动力辅助电源应用的电池安全标准》）。

1.19.6 Low Pressure Test 低气压试验

Under the condition of ambient temperature of 25℃ ±5℃, the battery is discharged in accordance with (1.8), and then charged in accordance with (1.7). The battery is placed in a low-pressure chamber. Adjust the air pressure in the test chamber to 11.6 KPa and the temperature to room temperature. Let it stand for 6 hours. Observe for 1 hour.

在环境温度 25℃±5℃的条件下，电池按照（1.8）进行标准放电，再按照（1.7）进行标准充电，电池放入低气压箱中，调节试验箱体中气压为 11.6KPa，温度为室温，静置 6h；观察 1h。

(Refer to UL1973-2022 Battery safety standards for stationary energy storage equipment and power-assisted power supply applications).

（参考 UL1973-2022《用于固定储能设备和动力辅助电源应用的电池安全标准》）。

1.19.7 Thermal Runaway Test 热失控试验

1) Battery cell state: Charged in accordance with (1.7).

1) 电芯状态：按照（1.7）进行标准充电；

2) Initial test environment: 25±5 ℃.

2) 测试初始环境：25±5℃；

3) Heating device: A heating film wraps the area except for the positive and negative electrodes and the safety valve.

3) 升温装置：加热膜包裹除正负极和安全阀外区域；

4) Temperature measuring device: K-type with specification of 24 or smaller, installed near the positive electrode, at the center of the large surface, between the heating film and the center of the large surface, etc. Continuously detect and record the surface temperature of the battery cell throughout the process.

4) 测温装置：K型24规格或更小，安装于正极附近、大面中心、加热膜与大面中心等位置，全程检测并记录电芯表面温度。

5) After the battery is installed, immediately start the heating device and control the temperature rise rate at 4-7 ℃/minute.

5) 电池安装完成后，立刻启动加热装置，控制温升速率在4~7℃/分钟；

6) When a voltage drop occurs in the battery and the surface temperature rise of the battery cell continuously and significantly exceeds the set temperature rise of the heating film, it is determined that thermal runaway has occurred. Record the temperature at which thermal runaway occurs.

6) 当电池产生电压降且电芯表面温升持续大幅度大于加热膜设定温升时，判定发生热失控，记录热失控发生温度；

7) If thermal runaway does not occur, use one of the following methods to promote the occurrence of thermal runaway:

7) 若热失控未发生，进行以下之一方法促使热失控发生：

1. Mechanical damage: such as puncture; 2. Overcharging, overdischarging or external short circuit; 3. Replace the heating source: such as an oven.

1. 机械破坏：例如穿刺 ；2. 过充电、过放电或外短路； 3. 更换加热源：例如烤箱。

(Refer to UL9540A Safety standard for test methods to assess thermal runaway fire propagation in battery energy storage systems).

（参考 UL9540A《评估电池储能系统中热失控火灾传播的测试方法的安全标准》）。

1.19.8 Nail Penetration Test 针刺试验

1) Under the condition of an ambient temperature of 25 ℃±2 ℃, the battery is discharged by a standard mode (1.8), then the battery is charged by a standard mode (1.7).

1) 在环境温度 25℃±2℃的条件下，电池按照（1.8）进行标准放电，再按照（1.7）进行标准充电。

2) Use a tungsten steel needle of 5 mm (the tip angle is 45°, and the surface of the needle is smooth, free of rust, oxide layer and oil stains). At a speed of 0.1 mm/s, make a shallow puncture of 30% perpendicularly to the direction of the battery cell plate. The steel needle stays in the battery cell.

2) 用 5mm 的钨钢针（针尖角度 45°，针的表面光洁，无锈蚀、氧化层及油污），以 0.1mm/s 的速度，从垂直于电芯极板的方向浅刺 30%，钢针停留在电芯中。

3) Observe for 1h.

3) 观察 1h。

(Refer to GB 44240-2024 Safety requirements for lithium batteries and battery packs for electrical energy storage systems).

（参考 GB 44240-2024 《电能存储系统用锂蓄电池和电池组安全要求》）。